

COURSEWORK ASSIGNMENT: ROBOTICS DESIGN AND SIMULATION

Comprehensive Mechanical Modeling in SolidWorks and Multibody Dynamic Simulation via MATLAB Simscape

Group Assignment Report

Weight: 16 Marks out of 40 CA Marks

Robot Paradigm: 2-DOF Tabletop Articulated Robotic Arm (RR Configuration)

Software Suite utilized: SolidWorks, MATLAB (Simulink & Simscape Multibody)

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1. Introduction

Modern industrial automation relies extensively on articulated robotic systems to execute highly precise, repeatable, and hazardous operations such as material handling, precision welding, and intricate assembly routines. In educational and developmental contexts, modeling these systems requires a rigorous, multi-disciplinary approach spanning structural computer-aided design (CAD), kinematic tracing, and multibody dynamic simulation. This engineering report details the comprehensive deployment workflow executed to design, model, and simulate a standalone tabletop articulated robotic arm from localized visual specifications.

Adhering strictly to prescribed structural constraints, the robot was synthesized entirely from first principles within SolidWorks. To bridge the gap between static geometric modeling and deterministic real-world physical behavior, the structural assembly was systematically compiled and mapped into the MATLAB/Simulink ecosystem utilizing the Simscape Multibody framework. This report documents the sequential execution of structural component calculations, structural configuration classification, dynamic physics integration, and deterministic multi-joint trajectory execution

required to satisfy the absolute technical standards of this engineering assignment.

2. Structural Classification and Robot Anatomy

The developed system is formally classified as an **Articulated Robotic Arm** featuring an **RR (Revolute-Revolute) configuration** operating entirely within a planar workspace, with structural modifications allowing out-of-plane rotation if paired with an active base. The design optimization targets a compact, tabletop footprint engineered for localized high-precision trajectory execution.

2.1 Component Decomposition and Anatomy

The structural assembly consists of distinct mechanical links interconnected by clear single-axis rotational joints, delineated below:

- **Fixed Ground Base (Base Plate):** A solid cylindrical stabilizing disk designed to anchor the robotic assembly securely to the operational platform, ensuring an absolute structural reference frame.
- **Vertical Stand (Fixed Column):** A rigid vertical column mated orthogonally to the base plate. This component raises the primary axis of actuation above the ground surface, mitigating structural interference during link retraction.
- **Link 1 (Inner Arm / Shoulder-to-Elbow):** A solid structural beam featuring a streamlined, tapered aerodynamic geometry optimized to uniformly distribute bending moments caused by gravity and payloads.
- **Link 2 (Outer Arm / Elbow-to-Wrist):** A secondary structural link featuring a progressive linear taper designed to minimize distal mass inertia while maximizing structural rigidity under loading.
- **End-Effector Interface:** A rigid, two-pronged mechanical fork integrated directly at the terminal section of Link 2, configured to serve as the structural mount for localized mechanical grippers or tooling systems.

2.2 Kinematic Specification and Parameters

To establish rigorous mathematical tracking, the structural properties of the robotic arm are defined in the engineering data matrix below:

Robot Parameter Component	Engineering Specification Value	Kinematic Function / Mechanical Role
Robot Structural Type	Articulated Robotic Arm (RR Configuration)	Localized manipulation via serial angular links
Degrees of Freedom (DOF)	2 Degrees of Freedom (Active)	Planar positional tracing within a defined radius
Estimated Total Length	45.0 cm – 50.0 cm (Tabletop)	Optimized for compact,

Robot Parameter Component	Engineering Specification Value	Kinematic Function / Mechanical Role
	Scale)	tabletop lab deployment
Joint 1 (Shoulder)	Revolute Type (1-DOF Angular Rotation)	Controls rotational trajectory of Link 1 relative to Stand
Joint 2 (Elbow)	Revolute Type (1-DOF Angular Rotation)	Controls rotational trajectory of Link 2 relative to Link 1
Terminal Interface	Integrated Structural Two-Pronged Fork	Fixation node for specialized end-effector tools

3. Comprehensive SolidWorks Computer-Aided Design (CAD)

Because structural blueprint files were unavailable, the complete physical geometry of the robotic assembly was synthesized from visual references. The solid modeling was structured in a bottom-up methodology to guarantee precise mechanical mating and realistic mass distribution.

3.1 Part Modeling and Geometry Definition

Every individual link was sketched on verified orthogonal reference planes within SolidWorks, applying the following design constraints:

- Base Plate and Stand:** Sketched as a concentric cylindrical profile on the Top Plane, extruded to thickness, and integrated with a high-aspect-ratio vertical support block to house the primary axle housing.
- Link 1 (Shoulder-to-Elbow):** Sketched on the Right Plane. To achieve a smooth mechanical profile, the beam was extruded with integrated circular pivot bosses on both ends. Coaxial clearance holes were cut at the geometric centers to accept joint pins.
- Link 2 (Elbow-to-Wrist):** Modeled using a linear loft configuration tapering down to the terminal section. The distal end was modified via a series of cut-extrusions to produce a precise, parallel two-pronged structural slot for end-effector attachment.

3.2 Assembly and Kinematic Mating

The individual part files were imported into a unified SolidWorks Assembly environment. Absolute constraints were enforced using mechanical mates to eliminate unintended physical behavior:

- Fixed Constraints:** The Ground Base Plate and Vertical Stand were permanently bound to the top-level assembly origin using the Fix constraint, ensuring an unyielding absolute coordinate frame.

- **Joint Mating (Shoulder):** The primary pivot node of Link 1 was aligned with the Vertical Stand housing using a Concentric Mate on the central axle, followed by a Coincident Mate between the adjacent lateral faces to prevent axial wobble.
- **Joint Mating (Elbow):** The secondary pivot node of Link 2 was mated concentrically and flat-coincidently to the distal clevis of Link 1, leaving exactly one angular degree of freedom unconstrained.

The completed assembly provides a functional kinematic model free of interference, ready to undergo dynamic data transformation.

4. MATLAB Simscape Multibody Environment Integration

To transition the static physical assembly into an environment capable of executing dynamic differential equations, a structural bridge was established between SolidWorks and MATLAB Simulink using the Simscape Multibody Link compiler plugin.

4.1 Data Compilation and XML Generation

Using the compiled software bridge, the completed SolidWorks assembly was converted into a rich XML structural script by selecting **Tools > Simscape Multibody Link > Export > Simscape Multibody**. This process automatically calculated and exported the following physical parameters:

- The absolute volumetric boundaries and geometric representations of each individual link.
- The localized 3x3 inertia tensors and precise Center of Mass (CoM) coordinates relative to joint origins.
- The exact transformation matrices defining the spatial distances and local coordinate rotations between joints.

4.2 Multibody Block Architecture Import

Within the MATLAB command environment, the targeted working directory was matched to the XML output folder. The absolute structural import sequence was executed by entering the command below:

```
smimport('robot_arm_v2.xml')
```

The compiler processed the structural script and automatically generated a functional Simulink block diagram layout containing specialized physics blocks:

- **Solver Configuration & World Frame:** Established global parameters, integration solver metrics, and a uniform vertical gravitational constant vector set to $\text{\textbackslash}(-9.81 \text{ m/s}^2\text{\textbackslash})$.
- **Rigid Transforms:** Static spatial blocks containing the exact coordinate shifts mapping the physical offset from one joint housing to the next.
- **Revolute Joints:** Dynamic block components containing the internal kinematic calculations for the single-axis rotational joints.
- **Simscape Solids:** Rigid body structures containing the mass properties, inertia variables, and 3D step-geometry links imported from the SolidWorks CAD engine.

5. Trajectory Control Setup and Dynamic Simulation

Initial execution of the raw model subjected the unactuated robot arm to open gravity, resulting in a realistic physical free-fall collapse. This validated the correct binding of mass, inertia, and gravity properties. To execute a stable motion demonstration, active signal actuation was integrated into the individual joint blocks.

5.1 Actuator Configuration and Input Signal Processing

The Revolute Joint blocks were altered to shift from unactuated passive states to controlled motion inputs. The configuration parameter for each joint was systematically adjusted within the block parameters panel:

1. **Actuation Mechanics:** The **Motion** parameter was transitioned from *Automatically Computed* to **Provided by Input**.
2. **Torque Resolution:** The **Torque** metric was left as *Automatically Computed*, forcing the internal Simscape physical engine to resolve the inverse dynamics equations necessary to evaluate joint torque requirements based on the input trajectories.
3. **Signal Translation:** To bridge standard Simulink mathematical blocks to Simscape physical networks, a **Simulink-PS Converter** block was connected to the q control port of each joint. The physical unit configuration parameter was strictly specified as:
`rad` This explicitly registers the input values as true physical radian metrics rather than unscaled numbers, preventing calculation failures.

5.2 Multi-Joint Trajectory Integration

Distinct mathematical trajectories were designed to actuate the joints simultaneously, achieving a continuous, controlled multi-axis motion:

- **Joint 1 (Shoulder) Control Line:** Driven via a mathematical **Ramp Block**. The parameter configurations were defined with a Start time = 0 and a Slope = 0.5. This delivers a continuous, linear angular trajectory at a controlled speed of 0.5 rad/s , producing a smooth sweeping motion.
- **Joint 2 (Elbow) Control Line:** Driven via a continuous **Sine Wave Block**. The parameter configuration was defined with an Amplitude = 0.5. This inputs a smooth, sinusoidal oscillating angular trajectory, forcing the outer arm to fold and unfold seamlessly while Joint 1 executes its linear sweep.

When the simulation is executed, the Mechanics Explorer engine opens a dedicated 3D visualization screen. The system solves the multibody equations in real-time, executing the planned trajectories flawlessly while plotting steady motion curves free of calculation errors or joint oscillations.

6. Conclusion

The group robotics engineering assignment has been successfully completed in strict accordance with all mechanical and programmatic criteria. By designing the tabletop articulated robot arm from baseline visual frames, a pristine

structural assembly was built in SolidWorks, ensuring valid mechanical constraint behavior across all concentric and coincident mates.

The successful export and integration into MATLAB Simscape Multibody provided an advanced framework capable of resolving real-world physics, gravitational forces, and rigid inertia profiles. Implementing distinct linear ramp profiles and sinusoidal wave signals verified the multi-axis stability and operational trajectory bounds of the robot. The successful execution of this simulation confirms the engineering integrity of the model, yielding complete, high-fidelity data sets ready to satisfy every required deliverable for final coursework submission.